Matthew Tang

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc) in Electrical Engineering — *Sep. 2026 – Apr. 2031*

- **Awards:** President's Scholarship of Distinction, Roberts Onsite Engineering Entrance Scholarship

Technical Skills

Languages: C++, Python, JavaScript, Java, HTML, CSS
Technologies: Firebase, Web Serial API, Progressive Web Apps, OpenCV, MediaPipe, UART/Serial, Power Budgeting
Tools: Altium, Autodesk Fusion, Git/GitHub, VS Code, Arduino IDE, 3D Printing, Oscilloscopes, Multimeters

Extracurriculars

Electrical Team Member — Sep. 2026 – Present
University of Waterloo Midnight Sun Solar Car Team — *Waterloo, ON*

- Designed the power tree for a multi-radio telemetry board (LoRa/FSK, Wi-Fi HaLow, 10BASE-T1S Ethernet, CAN-FD) streaming vehicle data from the solar car to a laptop, from 12 V input protection to 5 V and 3.3 V rails
- Built a component-level power budget from the datasheets of 7 devices (∼0.5 A on 3.3 V), then replaced the planned 5 V-to-3.3 V LDO with a second buck converter, cutting worst-case regulator heat loss ∼5× (0.85 W to 0.18 W)
- Defined the board's low-power strategy by evaluating each device's sleep states, load switches, and eFuses against the 8 W LV budget, reducing power lost by **78%** and keeping only hardware that improves fault recovery
- Captured a hierarchical Altium schematic with 14 V TVS input protection and 36 V-rated buck converters
- Onboarded by designing a 10–15 V to 5 V buck converter PCB in Altium, cutting noise and losses with copper pours and via-stitched ground planes, bench-tested at 90% efficiency and under 30 mV ripple at 1 A

DECA Executive Entrepreneurship Trainer — Sep. 2024 – June 2026
Earl Haig Secondary School — *Toronto, ON*

- Designed and delivered a structured DECA training program (lesson plans, case studies) for 10 - 20 members
- Hosted direct feedback sessions before competition, resulting in 12 provincial qualifiers and 3 international qualifiers
- Represented Ontario at DECA's international competition, Anaheim (2024) and California (2025)

Projects

Custom Pomodoro Timer | *C++, Arduino, JavaScript, Firebase, 3D Printing, Autodesk Fusion*

- Engineered a Pomodoro timer on an Arduino Uno R3 with a 16x2 LCD, push buttons and a buzzer, supporting a 25/5 auto mode and customizable focus length, break length and cycle count, with pause, resume and exit
- Programmed timer logic and UI in C++ as a non-blocking `millis()` state machine, handling state transitions, debounced button inputs, audio/visual alerts and an EEPROM-saved mute setting that survives power cycles
- Designed and 3D-printed a custom case in Autodesk Fusion, with a magnet-secured design for easy internal access
- Built Pomegranate, an installable web app controlling the timer over USB via a custom 115200-baud serial protocol, with task tracking, a daily journal and Firebase-synced analytics, increasing my productivity by **23%**

BetaVision | *Python, OpenCV, MediaPipe*

- Built a Python computer vision tool analyzing climbing videos with MediaPipe and OpenCV, tracking 16 joints per frame at 30 FPS and auto-annotating holds, starts, and top-outs with **95%** accuracy across 40+ gym videos
- Built CIELAB colour segmentation with lighting correction, auto-detecting 92% of route holds and volumes
- Engineered limb-to-hold contact detection using point-to-contour distance on 4 limbs, reconstructing move sequences with 90% accuracy, flagging off-route holds, and detecting top-outs within 0.2 s of the true finish
- Designed an OpenCV interface with zoom, scrolling, and auto/manual hold marking, cutting route setup by 80%

Experience

Camp Leader — July 2024 – Aug. 2026
City of Toronto — *Toronto, ON*

- Delivered 200+ hours of hands-on programming for groups of up to 50 children ages 6-9, planning and leading activities such as a bridge-building design challenge to foster creativity, teamwork, and problem-solving skills
- Conceived and led a cultural-awareness week, designing a Jeopardy-style trivia game, and organizing a multicultural potluck with 50+ participating families, while managing food-allergy compliance and safety
- Onboarded new staff through structured orientations and peer shadowing, raising positive parent feedback by **37%**